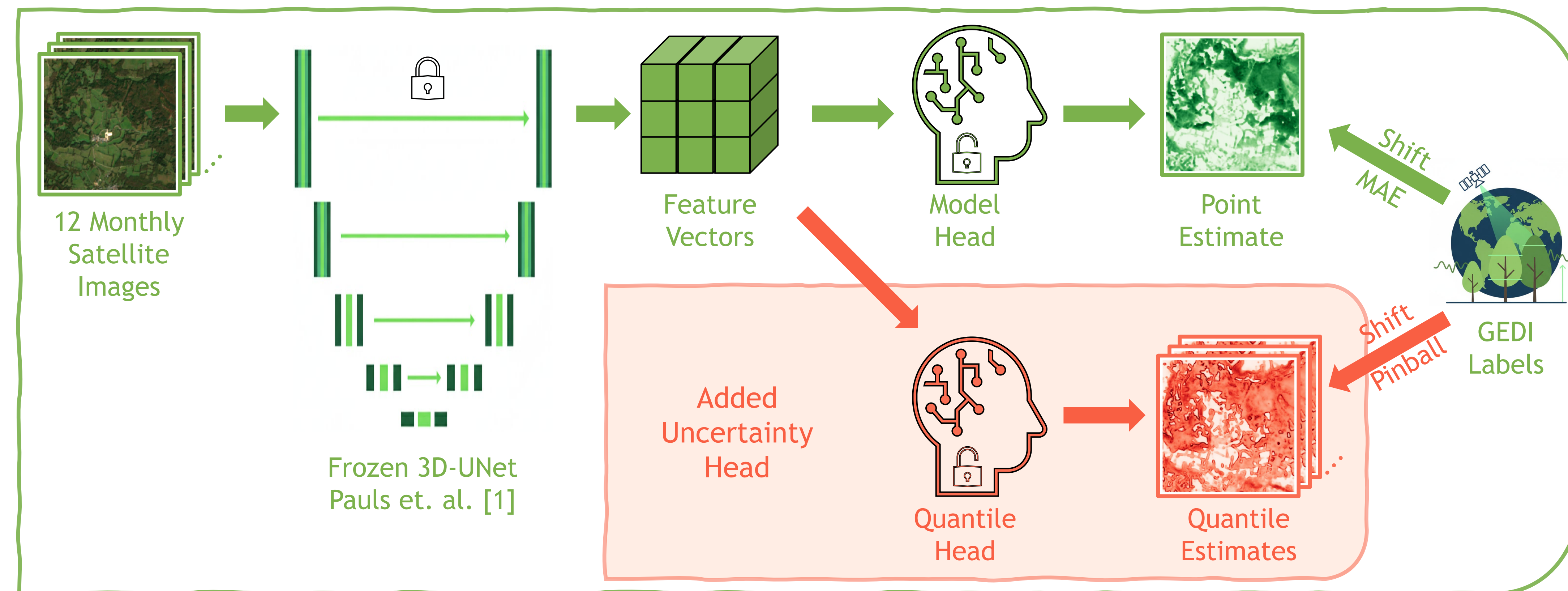


Why Tree Height Prediction?



Risk-sensitive downstream task need **Uncertainty Quantification**

Uncertainty Head



Pinball Loss

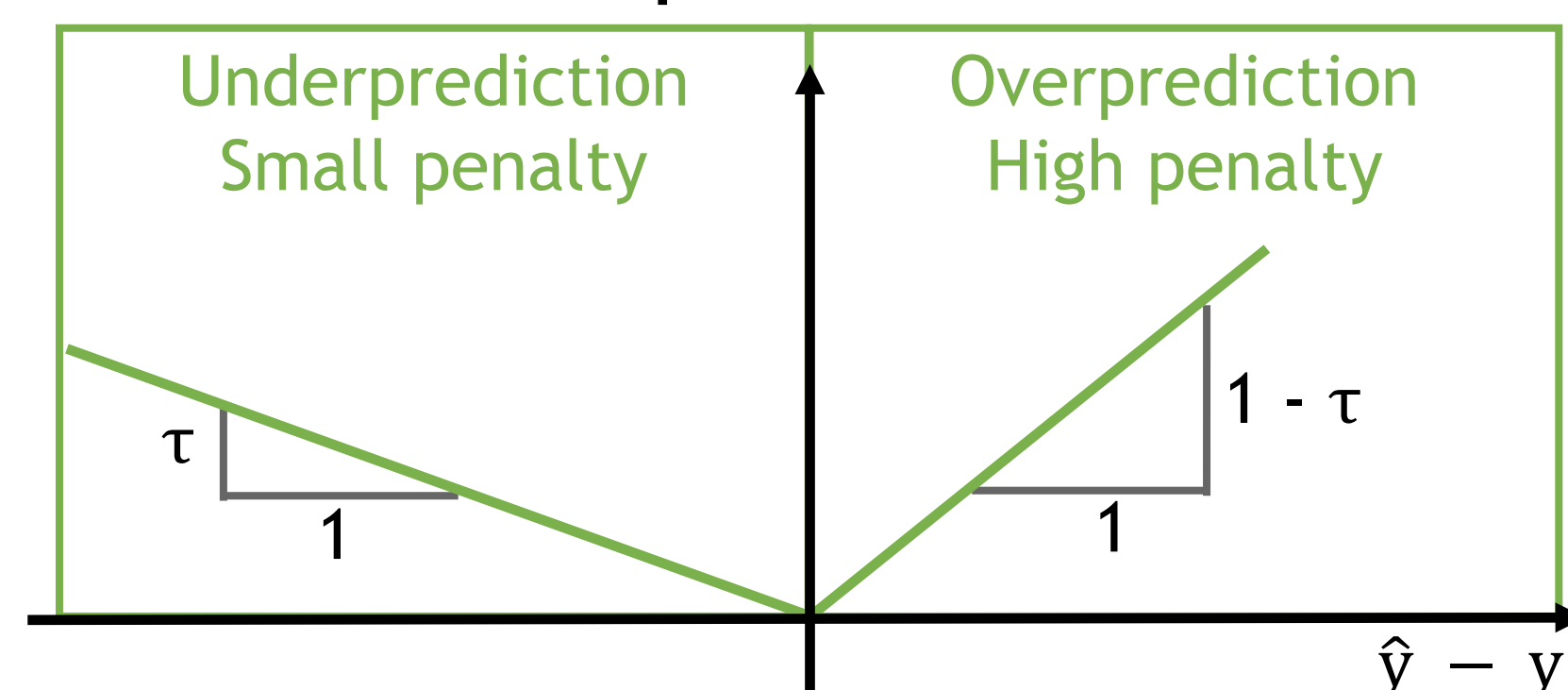
The Pinball loss is defined as

$$\mathcal{L}_\tau(y, \hat{y}) := \begin{cases} \tau \cdot (y - \hat{y}) & \text{if } \hat{y} \leq y \text{ (underprediction)} \\ (1 - \tau) \cdot (\hat{y} - y) & \text{if } \hat{y} > y \text{ (overprediction)} \end{cases}$$

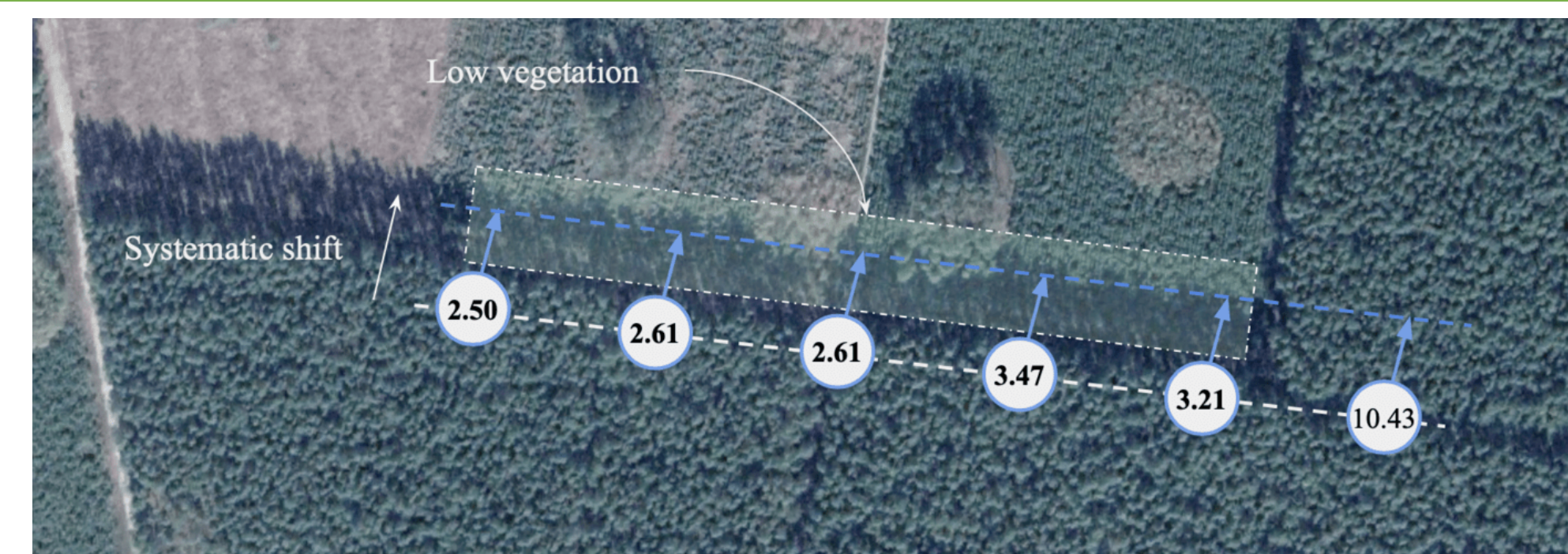
for a **target quantile** $\tau \in (0,1)$, the **ground truth value** $y \in \mathbb{R}$ and the **prediction** $\hat{y} \in \mathbb{R}$.

→ leads to prediction of conditional quantiles

Example: $\tau = 0.2$



Shift Loss^[2]



- Ground truth GEDI height measurements are taken from a sensor on the ISS (International Space Station)
- GEDI labels are sparsely available**
- GEDI labels are available on tracks
- All points of one track are approximately on one line
- GEDI labels are prone to **geolocation shifts**, which is approximately constant on tracks
- **Shift-resilient loss allows to shift all labels on a track by the same small offset to better fit the model's predictions**

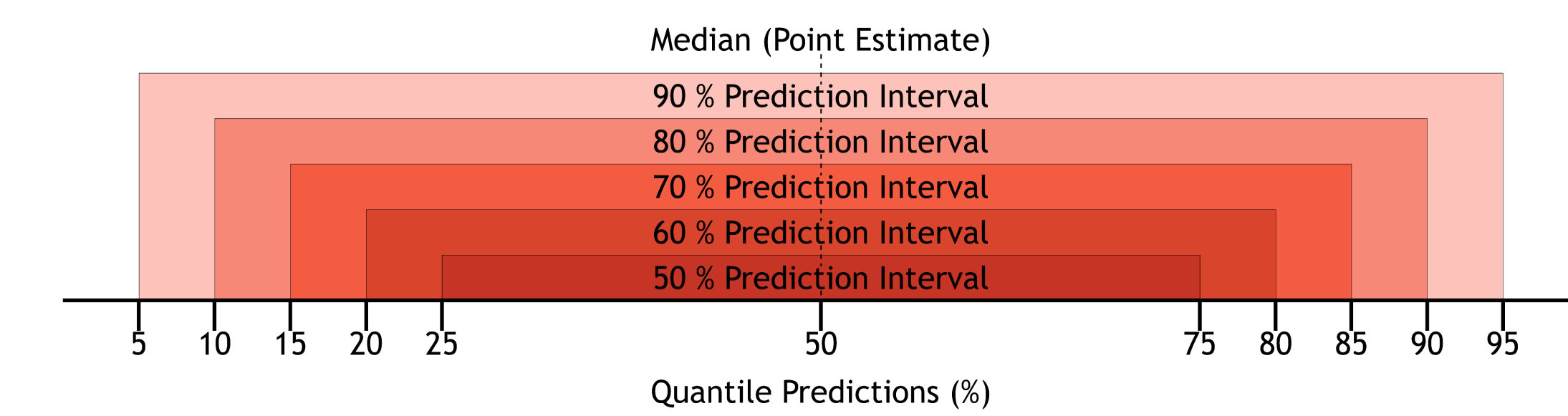
Experiments

- European Dataset - 10m resolution
 - Satellite images from Sentinel 1 & 2
 - Train heads on 1/4 of training data from Pauls et. al. [1]
 - Quantile head consists of two 3x3 convolutional blocks on 64 channels and a final 1x1 convolution
- Comparison models:**
- Lang et. al. [3]: Ensemble with Gaussian NLL loss model
 - Gaussian NLL loss head trained on our dataset
 - Log-Gaussian NLL loss head trained on our dataset

Prediction Intervals

Trained quantiles:

$\tau = (5\%, 10\%, 15\%, 20\%, 25\%, 50\%, 75\%, 80\%, 85\%, 90\%, 95\%)$



Quantitative Results

Metric	Mean Prediction Interval Width					Prediction Interval Coverage Probability				
Confidence Level	50%	60%	70%	80%	90%	50%	60%	70%	80%	90%
Lang et. al. [3]	5.54	6.87	8.39	10.22	12.76	37%	45%	52%	58%	64%
Gaussian w/ Shift	3.79	4.72	5.82	7.19	9.23	66%	75%	82%	88%	93%
Log-Gaussian w/ Shift	3.35	4.22	5.25	6.59	8.73	58%	69%	79%	86%	92%
Ours wo/ Shift	2.69	3.34	4.15	5.25	7.1	49%	59%	69%	79%	90%
Ours	2.43	3.01	3.72	4.68	6.34	47%	56%	66%	76%	87%

Table: Comparison of Mean Prediction Interval Width and Prediction Interval Coverage Probability

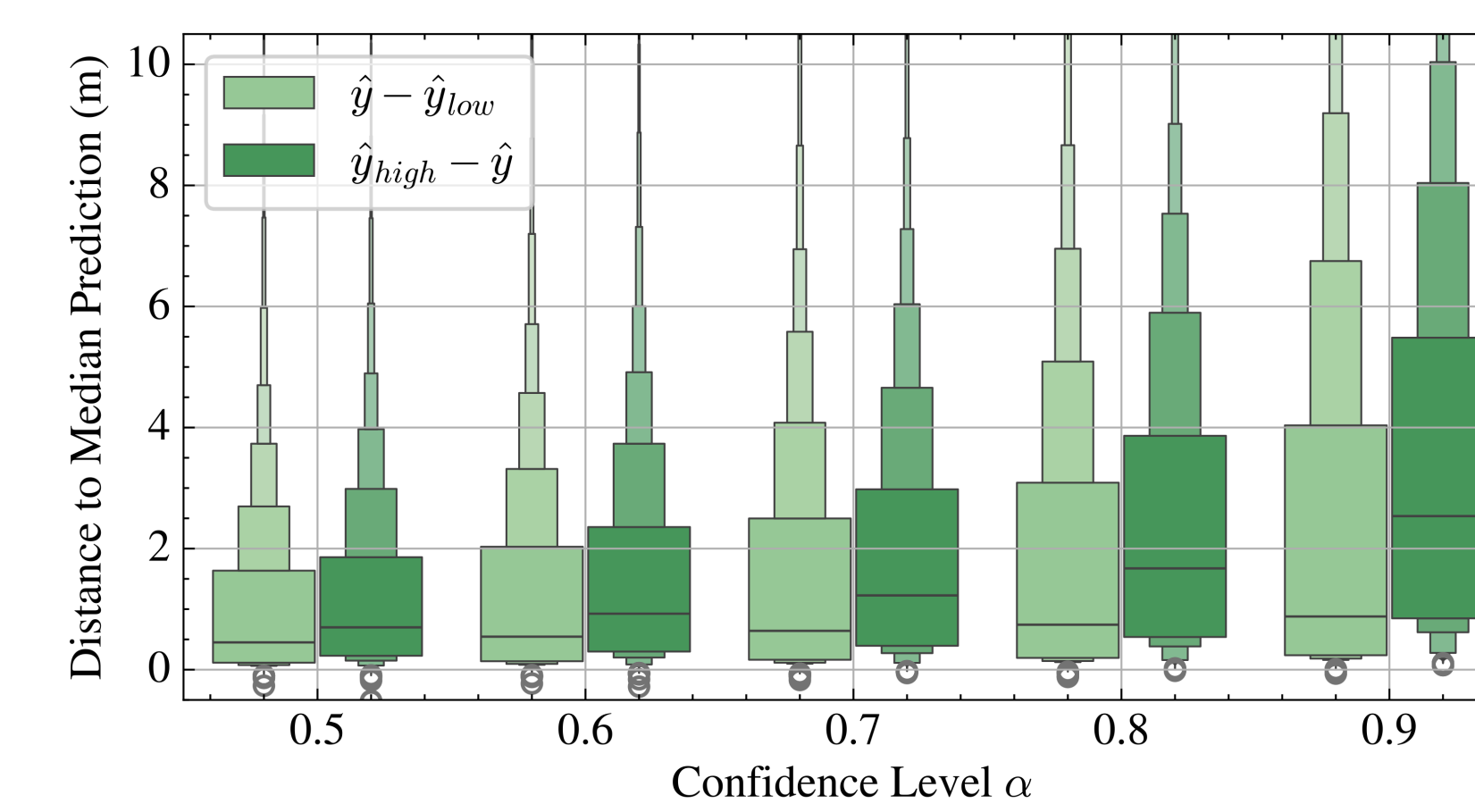


Figure: Boxplot of distance of prediction intervals $[\hat{y}_{low}, \hat{y}_{high}]$ boundaries to point-estimate, i.e. median prediction.

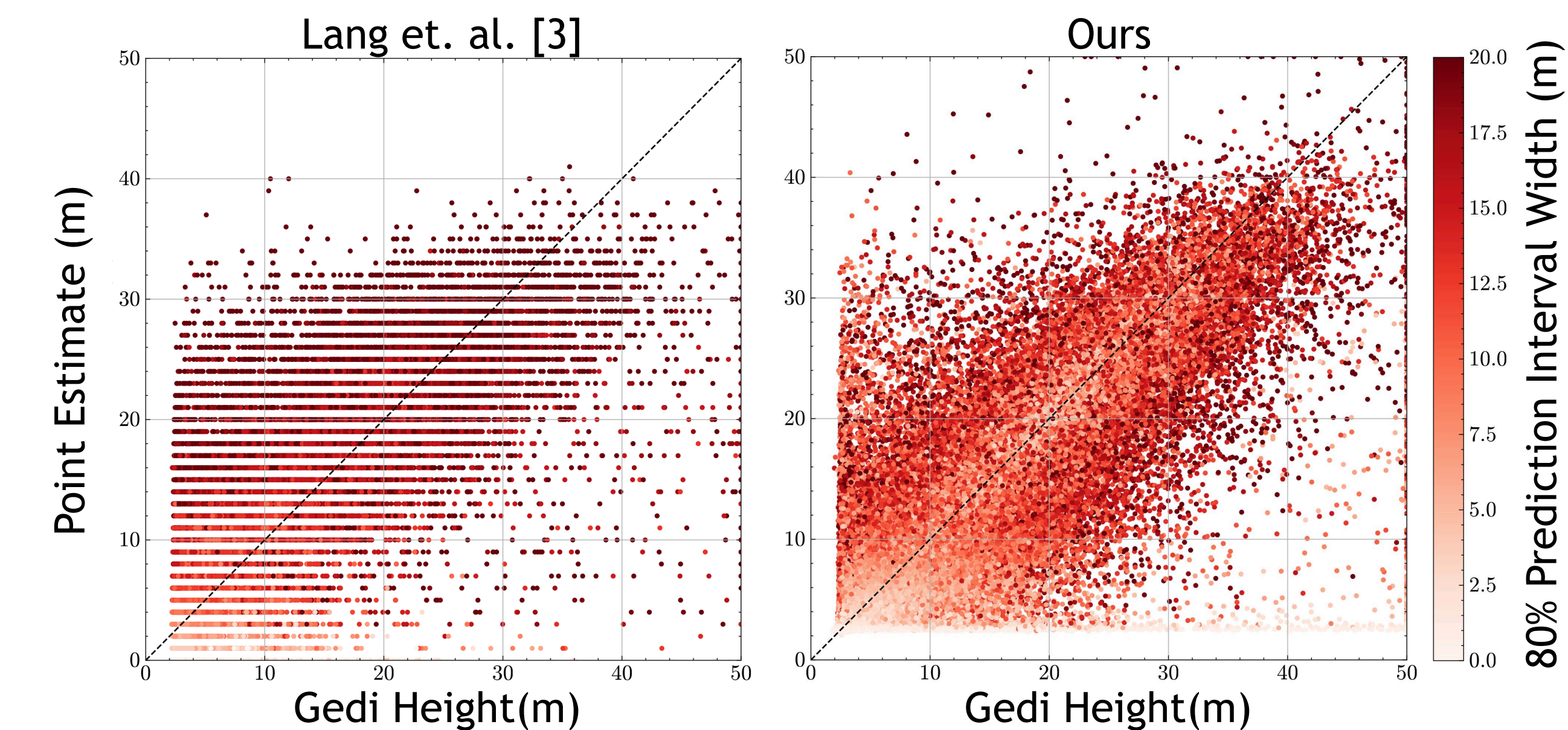


Figure: Scatter plot of GEDI height against predicted height with color dependent on the prediction interval width

Qualitative Results



Conclusion

- Higher uncertainty at forest border and in mountainous regions
- Shift-resilient loss reduces uncertainty and sharpens prediction
- Underprediction of tall trees remains a challenge
- Idea for future work: Use uncertainty quantification to filter faulty GEDI labels

